NOVA SCHOOL OF BUSINESS & ECONOMICS

Economics of education

3rd class

Outline for today

Why do governments want to intervene in education?

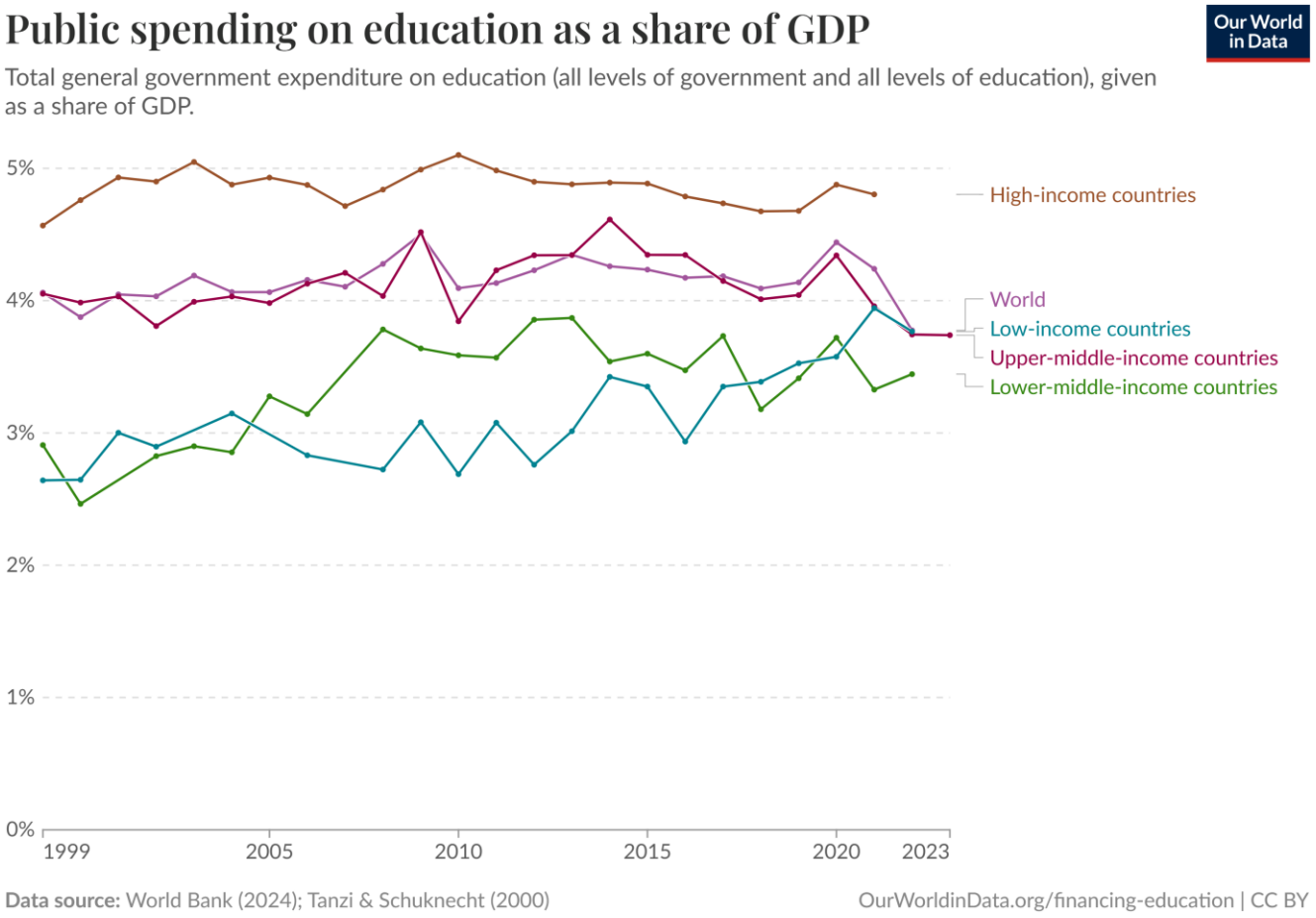
- The theory
- The empirical evidence

Readings

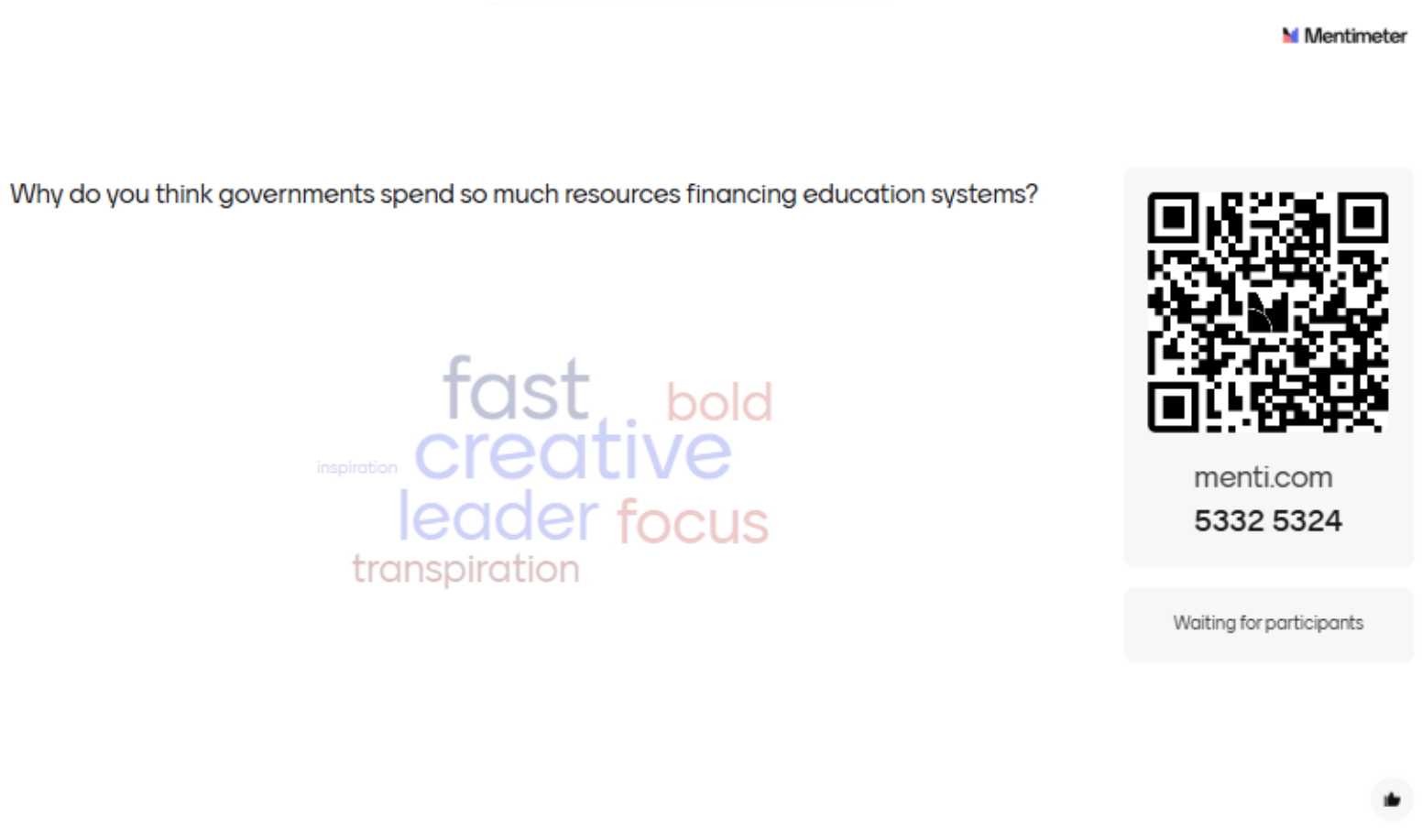
- Bradley & Green (2020) Chapter 32 - The economic role of the state in education
- Hanushek et al., (2011) Chapter 2 – Nonproduction Benefits of Education: Crime, Health, and Good Citizenship

Governmental intervention in education

- It is common to spend a large share of the Country's resources financing the education system



Quizz


Mentimeter
Why do you think governments spend so much resources financing education systems?
fast
bold
inspiration
creative
leader
focus
transpiration
menti.com
5332 5324
Waiting for participants

RP
Menti
Economics of educatio...
Choose a slide to present

Why do governments want to intervene in education?

Economic answer

- To generate economic growth
- State power (military power, bureaucratic capacity, technological leadership)

Why do governments want to intervene in education?

Modern economic answer

- To reduce crime
- To reduce health care costs
- To reduce welfare costs
- To fulfil basic human rights
- To increase civil engagement
- To decrease inequality
- To improve the life of its citizens (if they are benevolent)

Why do governments want to intervene in education?

- There may be advantages for the country
- By why is there a need to control and finance it?
- Education is a private good, with private benefits
- Why not let markets work?

Why do governments want to intervene in education?

Pareto efficient private market provision happens when (1st welfare theorem):

- Complete markets (all risks insurable; state-contingent contracts)
- No externalities
- Perfect information
- No market power/perfect competition
- Well-defined property rights

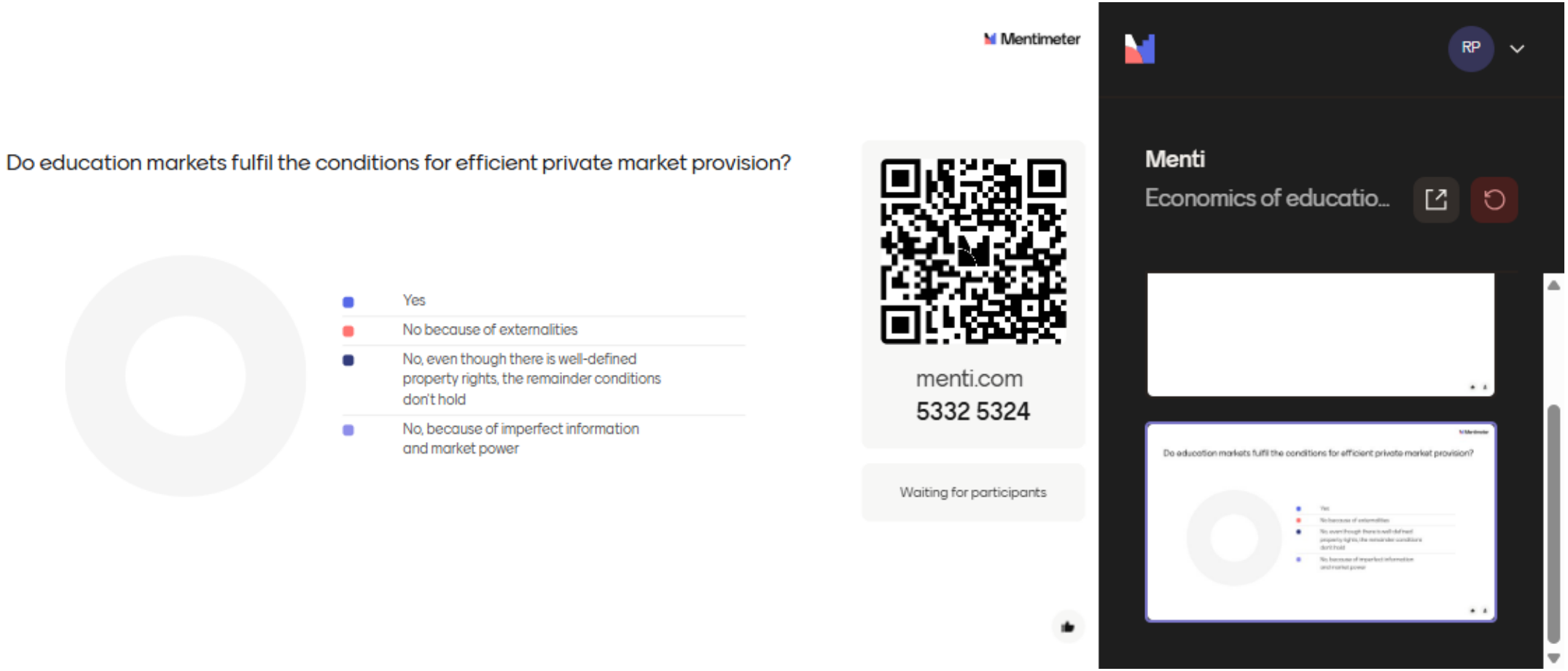
Mas-Colell, Whinston, and Green (1995), *Microeconomic Theory*, Oxford University Press

Why do governments want to intervene in education?

If all the conditions before apply, then a pareto efficient can be achieved through private markets if initial endowment redistribution is made appropriately (2nd welfare theorem)

Mas-Colell, Whinston, and Green (1995), *Microeconomic Theory*, Oxford University Press

Quizz


Mentimeter
Do education markets fulfil the conditions for efficient private market provision?
Yes
No because of externalities
No, even though there is well-defined property rights, the remainder conditions don't hold
No, because of imperfect information and market power
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Why do governments want to intervene in education?

Education is a good with well-defined property rights because:

- It is rival: marginal cost associated with educating each additional child
- It is excludable: access is easily controlled

Why do governments want to intervene in education?

Education markets have the potential to lead to market power due to economies of scale:

- Economies of scale arise when producers' average total cost falls as output increases
- Cost of library, gym, and science lab is high for few students, decreases as number of students increase

Why do governments want to intervene in education?

Education markets are of imperfect information since parents can't fully observe the quality of education:

- Education is a complex bundle of services
- Involves a large commitment of time
- Has a cumulative instructional process
- Uncertain future payoffs

Why do governments want to intervene in education?

Education is likely to produce externalities:

- Education likely has benefits beyond the private one
- Individuals will only take into consideration private costs and choose a suboptimal level of education

Why do governments want to intervene in education?

Difficult to collateralize loans for primary and secondary schooling and parents may face credit constraints:

- Insurance markets won’t work well
- Credit markets won’t either (that is also why there are usually governmental loans for higher education)

Why do governments want to intervene in education?

Unequal wealth distribution in society might condition optimal choices:

- Government might want to intervene to decrease financial constraints of access to schooling

Why do governments want to intervene in education?

Pareto efficient private market provision happens when (1st welfare theorem):

- Complete markets (all risks insurable; state-contingent contracts) ✗
- No externalities ✗
- Perfect information ✗
- No market power/perfect competition ✗
- Well-defined property rights ✓

Why do governments want to intervene in education?

The theory:

Gains at the societal level

- Financial benefits
- Non-monetary benefits
- State power

Plus inefficient private provision:

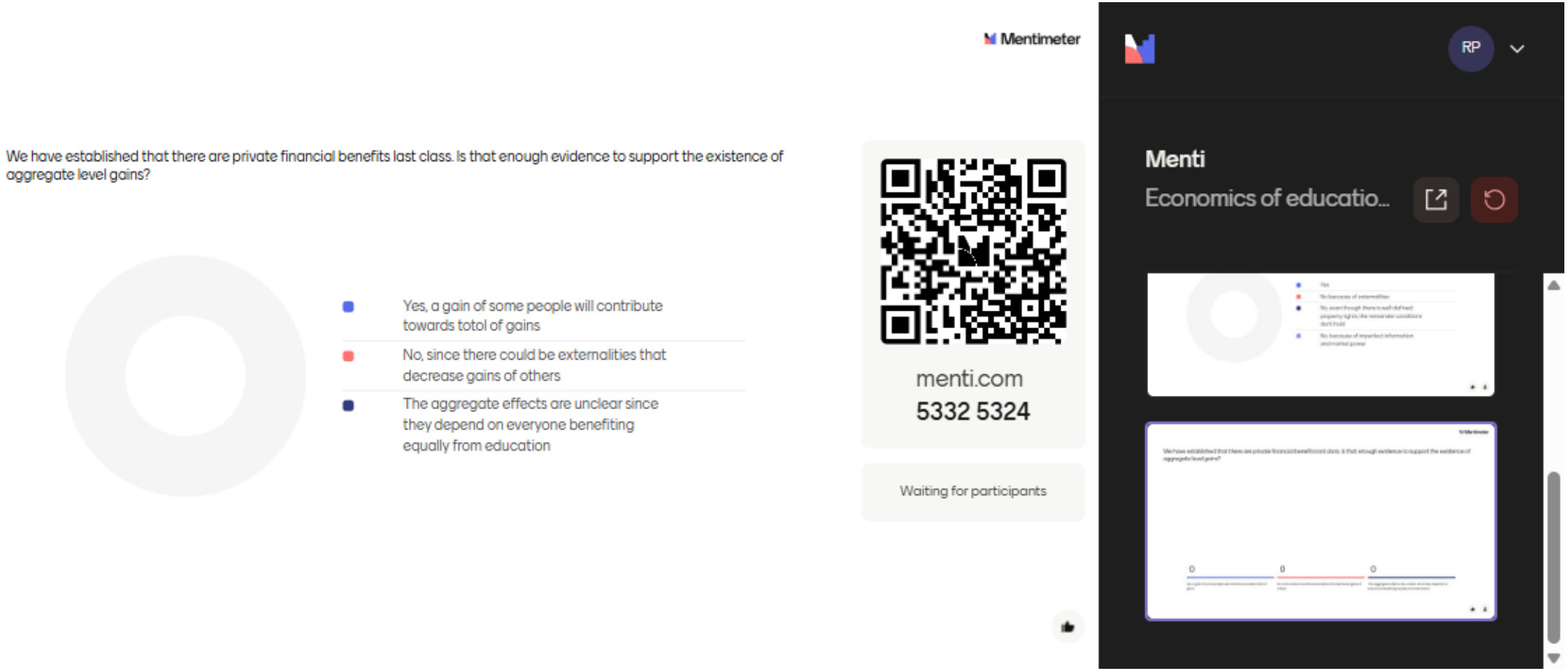
- Incomplete markets , externalities, incomplete information, economies of scale, unequal wealth distribution

Questions? Comments?

Why do governments want to intervene in education?

- Financial benefits
- Other benefits: crime, welfare, civil participation

Quizz


Mentimeter
We have established that there are private financial benefits last class. Is that enough evidence to support the existence of aggregate level gains?
Yes, a gain of some people will contribute towards totol of gains
No, since there could be externalities that decrease gains of others
The aggregate effects are unclear since they depend on everyone benefiting equally from education
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Why do governments want to intervene in education?

Financial benefits

There are (at least some) private gains to individuals! Isn’t this enough to establish causality?

- Spillovers / externalities (negative or positive)?
- Aggregate effects are unclear by looking at private gains

Why do governments want to intervene in education?

Financial benefits

- Let’s look at the literature that investigates **aggregate** gains of education
- We will also zoom in at instances where **externalities** are measured

Why do governments want to intervene in education?

Financial benefits

- There is a large substantial association between years of education and economic growth

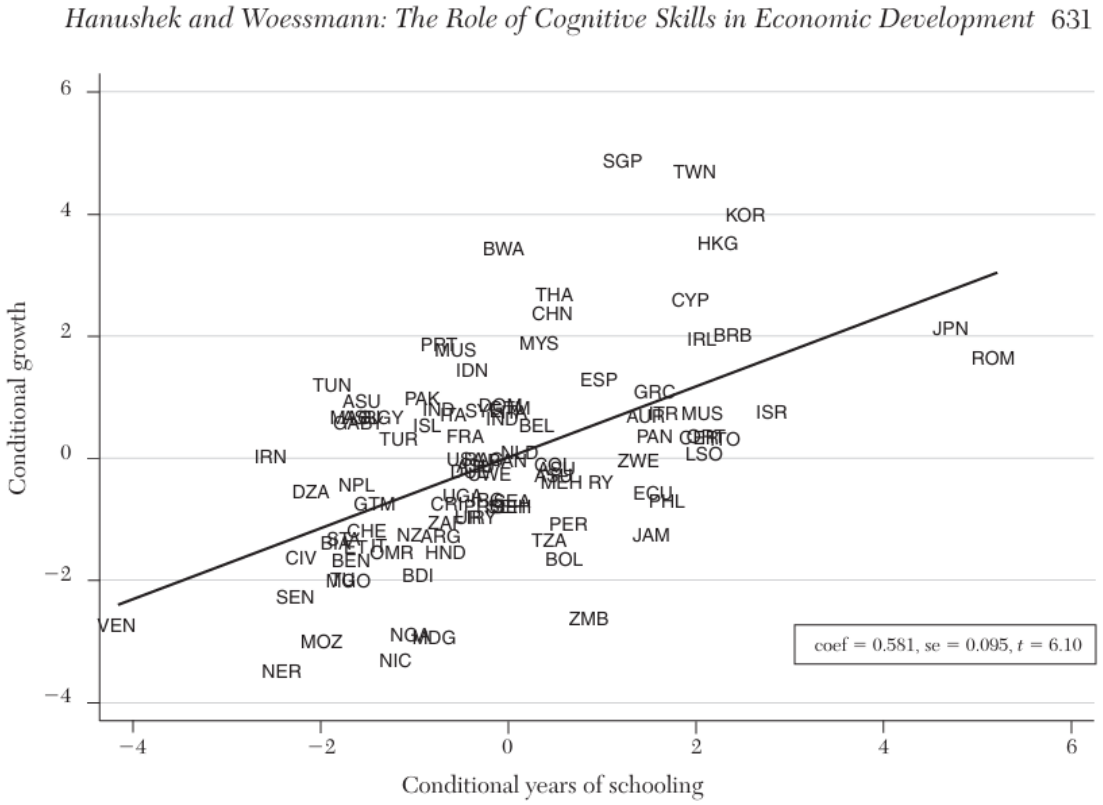

Figure 5. Added-variable Plot of Growth and Years of Schooling without Test Score Controls

Notes: Added-variable plot of a regression of the average annual rate of growth (in percent) of real GDP per capita in 1960–2000 on average years of schooling in 1960 and the initial level of real GDP per capita in 1960. Own calculations.

Source: Hanushek, E. A., & Woessmann, L. (2008). The role of cognitive skills in economic development. *Journal of economic literature*, *46*(3), 607-668.

Why do governments want to intervene in education?

Financial benefits

- Controlling for cognitive skills dramatically decreases the correlation

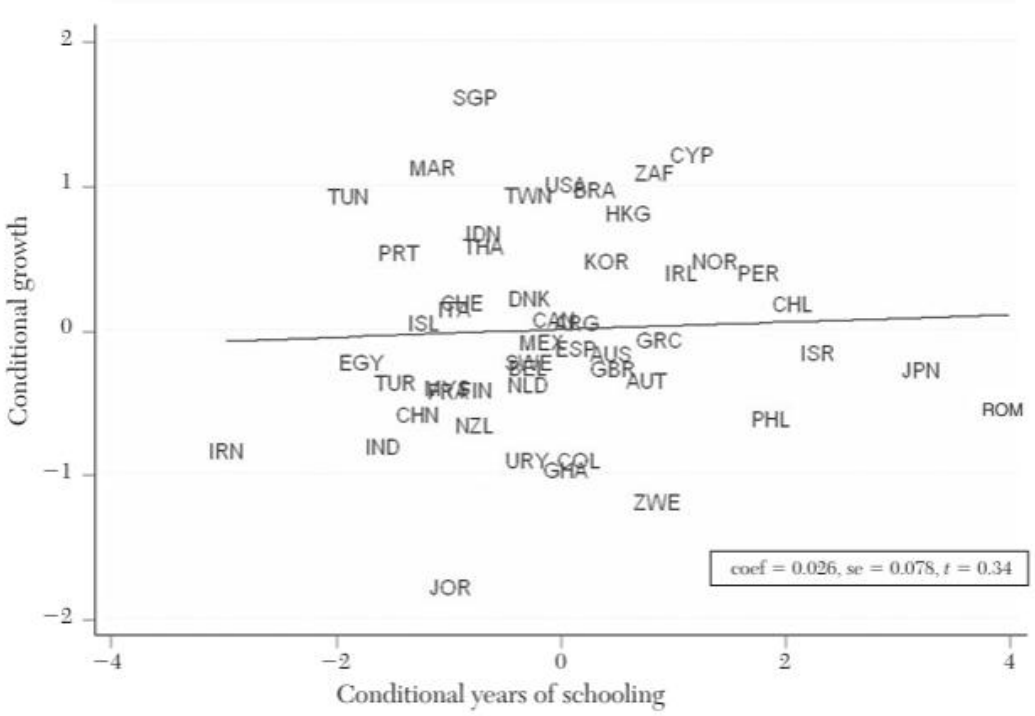

Figure 8. Added-Variable Plot of Growth and Years of Schooling with Test Score Controls

Notes: Added-variable plot of a regression of the average annual rate of growth (in percent) of real GDP per capita in 1960–2000 on the initial level of real GDP per capita in 1960, average test scores on international student achievement tests, and average years of schooling in 1960. Author calculations; see table 2, column 2.

Source: Hanushek, E. A., & Woessmann, L. (2008). The role of cognitive skills in economic development. *Journal of economic literature*, *46*(3), 607-668.

Why do governments want to intervene in education?

Financial benefits

- There is a large association between skills and economic growth
- Authors argue skills are the important factor, not education

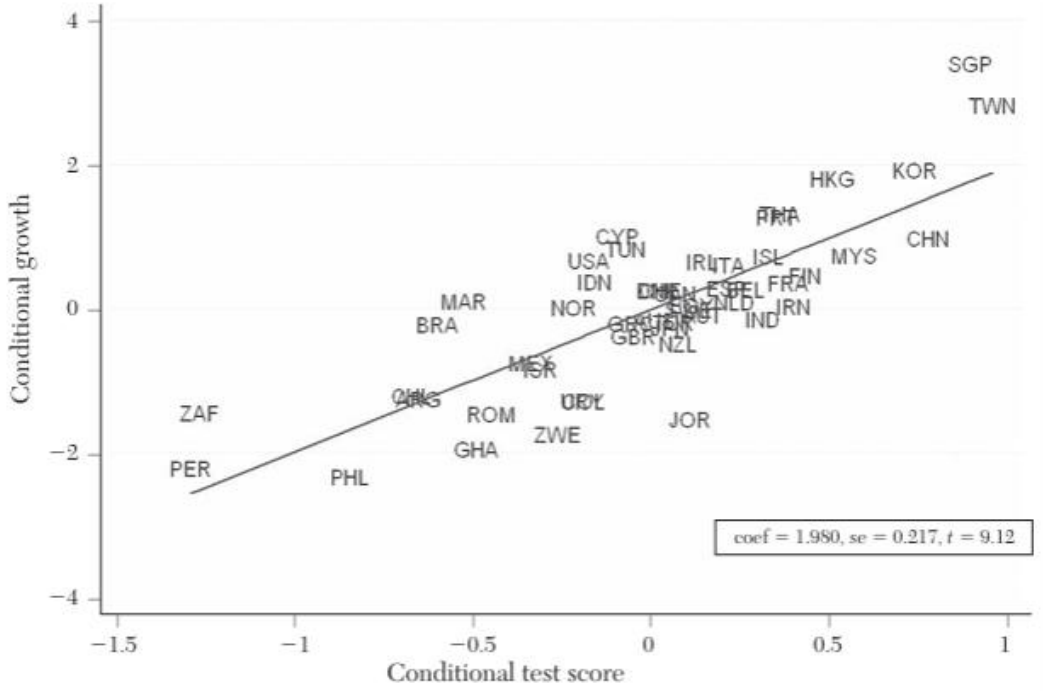

Figure 7. Added-Variable Plot of Growth and Test Scores

Notes: Added-variable plot of a regression of the average annual rate of growth (in percent) of real GDP per capita in 1960–2000 on the initial level of real GDP per capita in 1960, average test scores on international student achievement tests, and average years of schooling in 1960. Author calculations; see table 2, column 2.

Source: Hanushek, E. A., & Woessmann, L. (2008). The role of cognitive skills in economic development. *Journal of economic literature*, *46*(3), 607-668.

Why do governments want to intervene in education?

Financial benefits

- Trend in test scores (skills) predicts trend in growth
- Able to control for country fixed, but not dynamic, differences

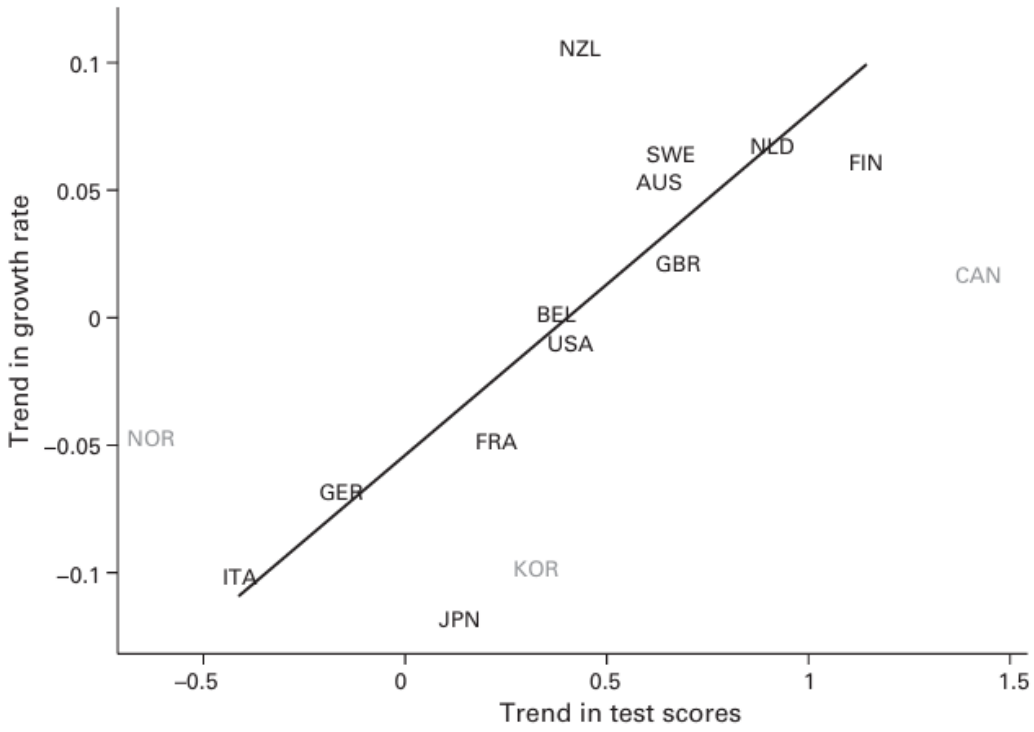

Figure 4.2

Trends in test scores and trends in growth rates

Notes: Scatter plot of trend in the growth rate of GDP per capita from 1975 to 2000 and trend in test scores, which is equivalent to the first column of table 4.2. Three countries without test scores before 1972 in light gray; regression line refers to the remaining twelve countries. See text for details.

Source: Hanushek, E. A., & Woessmann, L. (2008). The role of cognitive skills in economic development. *Journal of economic literature*, *46*(3), 607-668.

Why do governments want to intervene in education?

Financial benefits

- Association between cognitive skills and economic growth strong and stable across decades

Table 1 Years of schooling versus cognitive skills in growth regressions

	(1)	(2)	(3)	(4)[a]	(5)[b]	(6)[c]	(7)[d]	(8)[e]	(9)[f]
Cognitive skills		2.015	1.980	1.975	1.933	1.666	1.265	1.239	1.985
		(10.68)	*(9.12)*	*(8.28)*	*(8.29)*	*(5.09)*	*(4.06)*	*(4.12)*	*(7.83)*
Years of schooling 1960	0.369		0.026	0.024	0.025	0.047	0.004	−0.049	−0.090
	(3.23)		*(0.34)*	*(0.78)*	*(0.29)*	*(0.54)*	*(0.05)*	*(0.66)*	*(1.02)*
GDP per capita 1960	−0.379	−0.287	−0.302	−0.298	−0.298	−0.255	−0.351	−0.310	−0.879
	(4.24)	*(9.15)*	*(5.54)*	*(6.02)*	*(5.04)*	*(3.12)*	*(6.01)*	*(5.73)*	*(3.39)*
No. of countries	50	50	50	50	52	50	47	45	50
R^2 (adj.)	0.252	0.733	0.728	0.728		0.706	0.784	0.797	0.637

Notes Dependent variable: average annual growth rate in GDP per capita, 1960–2000. Regressions include a constant. Test scores are average of math and science, primary through end of secondary school, all years. Absolute t-statistics in parentheses

[a] Measure of years of schooling refers to the average between 1960 and 2000

[b] Robust regression including the two outliers of Botswana and Nigeria (using *rreg* robust estimation command implemented in Stata)

[c] Specification includes dummies for the eight world regions depicted in Fig. 1

[d] Specification includes additional controls for openness and property rights

[e] Specification includes additional controls for openness, property rights, fertility, and tropical location

[f] GDP per capita 1960 measured in logs

Source: Hanushek, E. A., & Woessmann, L. (2008). The role of cognitive skills in economic development. *Journal of economic literature*, *46*(3), 607-668.

Why do governments want to intervene in education?

Financial benefits

- Once you control for cognitive skills, years of schooling are irrelevant for explaining growth
- Authors argus that this is suggestive of causality

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Why do governments want to intervene in education?

Financial benefits

- Strong correlational evidence that suggests that skills are good predictors of economic growth
- Growth rates of education predict growth rates of GDP
- This excludes country specific static characteristics driving the relationship
- Some differences in dynamic events could still bias the results
- Results stable across decades (less room for he above)

Source: Hanushek, E. A., & Woessmann, L. (2008). The role of cognitive skills in economic development. *Journal of economic literature*, *46*(3), 607-668.

Why do governments want to intervene in education?

Financial benefits

- Strong suggestive evidence that cognitive skills might be one of the driving forces of economic development

Source: Hanushek, E. A., & Woessmann, L. (2008). The role of cognitive skills in economic development. *Journal of economic literature*, *46*(3), 607-668.

Why do governments want to intervene in education?

Financial benefits

- Study that includes 66 countries (1980-1999)
- Authors document that countries with higher educational growth experienced faster value-added and employment growth
- That growth happened in schooling intensive industries

Ciccone, A., & Papaioannou, E. (2009). Human capital, the structure of production, and growth. *The review of economics and statistics*, *91*(1), 66-82.

Why do governments want to intervene in education?

Financial benefits

- HCINT=industry-level schooling intensity
- ΔSCH=annual country-level change in average years of schooling over the 1980–1999 period

TABLE 3.—HUMAN CAPITAL LEVEL, HUMAN CAPITAL ACCUMULATION AND INDUSTRY GROWTH (ALL COUNTRIES AND COUNTRIES OPEN TO INTERNATIONAL TRADE ONLY)

	All Countries				Countries Open to International Trade			
	Value-Added Growth		Employment Growth		Value-Added Growth		Employment Growth	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Initial schooling interaction	0.0021	0.0020	0.0030	0.0027	0.0044	0.0043	0.0055	0.0052
[*SCH* × *HCINT*]	*(2.66)*	*(2.55)*	*(4.77)*	*(4.32)*	*(4.90)*	*(5.00)*	*(6.99)*	*(6.62)*
Schooling accumulation interaction	0.0794	0.0760	0.1222	0.1018	0.1661	0.1681	0.1844	0.1752
[Δ*SCH* × *HCINT*]	*(2.43)*	*(2.28)*	*(4.43)*	*(3.57)*	*(4.58)*	*(4.69)*	*(5.61)*	*(5.39)*
All controls	No	Yes	No	Yes	No	Yes	No	Yes
Initial conditions	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R^2	0.461	0.465	0.518	0.532	0.568	0.569	0.654	0.674
Observations	1,049	964	1,134	998	523	523	553	530
Countries	46	40	47	41	21	21	22	21
Country fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

The dependent variable in columns 1–2 and 5–6 is the annual growth rate of value added at the country-industry level for the period 1980–1999. The dependent variable in columns 3–4 and 7–8 is the annual growth rate of employment over the same period. All value-added growth models include the initial log of value added and all employment growth models include the initial log employment at the country-industry level (coefficients not reported). Columns 1–4 are estimated for the largest possible sample. Columns 5–8 restrict estimation to countries that have been open to international trade since 1970, according to Sachs and Warner (1995) criterion of openness as updated and extended by Wacziarg and Welch (2003). The initial schooling interaction is country-level average years of schooling in 1980 (*SCH80*) multiplied by industry-level schooling intensity (*HCINT*). The schooling accumulation interaction is the product of the industry-level schooling intensity (*HCINT*) with the annual country-level change in average years of schooling over the 1980–1999 period (Δ*SCH*). In even columns we jointly control for the physical capital interaction (*CAPINT* × *K/Y*), the rule of law interaction (*RLAW* × *CONTRACT*), and the finance interaction (*PRIV* × *EXTFIN*). (These specifications are analogous to table 2, columns 11 and 12.) All specifications also include country fixed effects and industry fixed effects (coefficients not reported). Absolute values of *t*-statistics based on robust (heteroskedasticity-adjusted) standard errors are reported in parentheses below the coefficients. The data appendix gives detailed variable definitions and data sources.

Ciccone, A., & Papaioannou, E. (2009). Human capital, the structure of production, and growth. *The review of economics and statistics*, *91*(1), 66-82.

Why do governments want to intervene in education?

Financial benefits

- Industries that captured the "educational growth" generated more value added, and more employment
- Especially if open to trade

TABLE 3.—HUMAN CAPITAL LEVEL, HUMAN CAPITAL ACCUMULATION AND INDUSTRY GROWTH (ALL COUNTRIES AND COUNTRIES OPEN TO INTERNATIONAL TRADE ONLY)

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All controls	No	Yes	No	Yes	No	Yes	No	Yes
Initial conditions	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R^2	0.461	0.465	0.518	0.532	0.568	0.569	0.654	0.674
Observations	1,049	964	1,134	998	523	523	553	530
Countries	46	40	47	41	21	21	22	21
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Why do governments want to intervene in education?

Financial benefits

- The industries that incorporated more higher educated individuals produced more growth and employment, particularly if countries were open to international trade
- Does not explain why some industries are schooling intensive
- Nevertheless, is evidence that gives a plausible explanation of the mechanism of education on economic growth

Ciccone, A., & Papaioannou, E. (2009). Human capital, the structure of production, and growth. *The review of economics and statistics*, *91*(1), 66-82.

Why do governments want to intervene in education?

Financial benefits

- US sample
- Instruments schooling decisions and looks for effects on private and social returns to education

Source: Acemoglu, D., & Angrist, J. (2000). How large are human-capital externalities? Evidence from compulsory schooling laws. *NBER macroeconomics annual*, *15*, 9-59.

Why do governments want to intervene in education?

Financial benefits

- Instruments are differences across state and years of compulsory schooling laws and child labour laws
- More restrictive labour laws implied more schooling

Source: Acemoglu, D., & Angrist, J. (2000). How large are human-capital externalities? Evidence from compulsory schooling laws. *NBER macroeconomics annual*, *15*, 9-59.

Table 3 DESCRIPTION OF CHILD LABOR AND COMPULSORY SCHOOLING LAWS

Year at Age 14 (Census Year)	*Earliest Dropout Age (1)*	*Latest Enrollment Age (2)*	*Minimum Schooling for Dropout (3)*	*Earliest Work Age (4)*	*Required Schooling for Work Permit (5)*
1914 (50)	15.31	7.49	1.90	11.00	1.70
	(1.20)	(0.52)	(3.40)	(5.75)	(2.56)
1917 (50)	15.55	7.63	1.93	13.43	2.98
	(0.89)	(0.49)	(2.74)	(1.98)	(2.66)
1921 (50)	15.69	7.42	4.28	13.94	4.19
	(0.99)	(0.51)	(3.63)	(1.71)	(2.97)
1924 (60)	15.88	7.29	5.64	14.11	4.91
	(0.97)	(0.57)	(3.64)	(1.33)	(3.04)
1929 (60)	15.97	7.30	5.66	14.16	5.31
	(0.93)	(0.58)	(3.62)	(1.33)	(3.01)
1935 (70)	15.96	7.24	7.24	14.14	6.02
	(0.94)	(0.55)	(3.73)	(0.76)	(2.67)
1939 (70)	16.16	7.16	7.29	14.15	6.01
	(1.05)	(0.51)	(3.74)	(0.77)	(2.70)
1946 (80)	16.31	7.09	7.91	14.77	4.67
	(0.63)	(0.53)	(4.00)	(1.16)	(3.37)
1950 (80)	16.27	7.08	7.94	15.03	3.51
	(0.60)	(0.53)	(4.49)	(1.14)	(3.47)
1954 (80)	16.30	7.05	7.79	15.02	4.06
	(0.63)	(0.52)	(4.65)	(1.20)	(3.67)
1959 (90)	16.25	7.05	7.40	15.19	3.49
	(0.60)	(0.53)	(4.79)	(1.19)	(3.56)
1964 (90)	16.20	7.05	7.44	15.17	3.51
	(0.60)	(0.54)	(4.79)	(1.22)	(3.57)

Notes: Standard deviations are in parentheses. All other entries are means. The data are from the Census IPUMS for 1950 through 1990, with the sample restricted to white men aged 40–49 in the Census year. See Appendix B for sources and method.

Why do governments want to intervene in education?

Using instruments

Not using instruments

Financial benefits

- Evidence for private and external returns

Table 7 2SLS ESTIMATES OF PRIVATE AND EXTERNAL RETURNS TO SCHOOLING–STATE-OF-BIRTH INSTRUMENTS, 1960–1980 AND MEN AGED 40–49

	Individual Schooling Endogenous				*Individual Schooling Exogenous*			
	(1)	*(2)*	*(3)*	*(4)*	*(5)*	*(6)*	*(7)*	*(8)*
Second-Stage Estimates								
Instrument set	QOB & CL	QOB & CA	QOB, CA & CL	QOB, CA & CL, interactions	CL	CA	CL & CA	QOB, CA & CL, interactions
Private return to schooling	0.074 (0.012)	0.074 (0.012)	0.075 (0.012)	0.060 (0.013)	0.073 (0.0003)	0.073 (0.0003)	0.073 (0.0003)	0.073 (0.0003)
External return to schooling	0.003 (0.040)	0.017 (0.043)	0.004 (0.035)	0.005 (0.033)	0.002 (0.038)	0.018 (0.042)	0.006 (0.033)	−0.011 (0.030)

Instrument affects own schooling (private returns) and average schooling in the state (external returns)

Source: Acemoglu, D., & Angrist, J. (2000). How large are human-capital externalities? Evidence from compulsory schooling laws. *NBER macroeconomics annual*, *15*, 9-59.

Why do governments want to intervene in education?

Financial benefits

- US sample
- First instrument of this paper is the age structure in 1970
- Placebo check

Table 6. Partial correlation between 1970 demographic structure and changes in population, labor force, inflows of domestic workers, inflow of foreign workers: Census data

	Change populat. 1980–1990 (1)	Change labor force 1980–1990 (2)	Change participat. rate 1980–1990 (3)	Net domestic inflow 1980–1990 (4)	Internat. inflow 1980–1990 (5)
Share of young, 1970	0	0	0	0	–
Share of middle-aged, 1970	0	0	0	0	+
Share of old, 1970	0	0	0	0	0

Notes: Each entry is a separate regression. Entries in the first column are +, 0 or − indicating whether the regression of percentage change in population on the share of a demographic group yields a positive, insignificant or negative coefficient. Entries in the remaining 4 columns are obtained similarly. Young individuals are aged 16–25; middle-age ones 26–60 and old ones 61–70.

Source: Moretti, E. (2004). Estimating the social return to higher education: evidence from longitudinal and repeated cross-sectional data. *Journal of econometrics*, *121*(1-2), 175-212.

Why do governments want to intervene in education?

Financial benefits

- US sample
- Second instrument of this paper is land grant
- Morril act (1862), gave every state that had remained in the Union a grant to establish colleges

Source: Moretti, E. (2004). Estimating the social return to higher education: evidence from longitudinal and repeated cross-sectional data. *Journal of econometrics*, *121*(1-2), 175-212.

Why do governments want to intervene in education?

Financial benefits

- Impact on education

Table 7
The effect of the presence of a land grant university on the distribution of education in city: 1990 Census data

	College + (1)	Some college (2)	High-school (3)	High-school drop out (4)
Land-grant	0.05	0.002	−0.02	−0.03
	(0.01)	(0.004)	(0.006)	(0.005)
R^2	0.39	0.64	0.62	0.57
Region effects	Yes	Yes	Yes	Yes
Unemployment and city controls	Yes	Yes	Yes	Yes

Notes: Each entry is a separate regression. The dependent variable is the percentage of individuals in city belonging to one of the four education groups. Standard errors in parenthesis.

Source: Moretti, E. (2004). Estimating the social return to higher education: evidence from longitudinal and repeated cross-sectional data. *Journal of econometrics*, *121*(1-2), 175-212.

Why do governments want to intervene in education?

Financial benefits

- Both instruments show positive externalities
- Positive externalities vary by education level
- Lower educated groups benefit proportionally more

Table 5
The effect of changes in share of college graduates on wage of education groups: 1980 and 1990 Census data

	1980–1990				1990		1980	
	OLS (1)	OLS (2)	2SLS (3)	2SLS (4)	OLS (5)	2SLS (6)	OLS (7)	2SLS (8)
First stage								
Age structure			3.36	3.52				
			(0.67)	(0.67)				
Land Grant						0.05		0.05
						(0.01)		(0.01)
Second stage								
Less than high-school	1.47	1.44	2.22	1.91	0.75	0.77	0.67	0.58
	(0.15)	(0.15)	(0.51)	(0.52)	(0.06)	(0.20)	(0.07)	(0.17)
High-school	1.38	1.34	2.08	1.67	0.85	0.84	0.74	0.74
	(0.13)	(0.13)	(0.45)	(0.45)	(0.06)	(0.18)	(0.06)	(0.14)
Some college	1.29	1.25	1.66	1.24	0.86	0.94	0.70	0.63
	(0.12)	(0.12)	(0.42)	(0.42)	(0.06)	(0.18)	(0.06)	(0.14)
College +	0.87	0.83	0.86	0.47	0.74	0.55	0.70	0.45
	(0.11)	(0.10)	(0.35)	(0.37)	(0.06)	(0.19)	(0.07)	(0.17)
City effects	Yes	Yes	Yes	Yes				
Region effects					Yes	Yes	Yes	Yes
Unempl. and other city controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Katz and Murphy Index		Yes		Yes				

Notes: Each entry is a separate regression. Years effects are included in all models. Dependent variables are education-city-time specific intercepts in a regression of log wages on sex, race, Hispanic origin, US citizenship and a quadratic term in potential experience. Entries are the coefficients on the percentage college graduates. Instrument in columns 3 and 4 is age structure. Instrument in columns 6 and 8 is land-grant college. Standard errors in parentheses.

Source: Moretti, E. (2004). Estimating the social return to higher education: evidence from longitudinal and repeated cross-sectional data. *Journal of econometrics*, *121*(1-2), 175-212.

Why do governments want to intervene in education?

Financial benefits

Recent strand of economic literature that looks at how religious spread accompanied by schooling lead to economic prosperity:

- Jesuits increased literary rates and schooling in 1600's in Argentina, Brazil, and Paraguay (1)
- Protestants influenced literacy in 1900's in Prussia (2)
- Papal schism led to three universities being funded in 1386 in Germany (3)

Why do governments want to intervene in education?

Financial benefits

Recent strand of historic economic literature that looks at how religion propelled schooling led to economic prosperity:

- Literature finds those coincide with higher skills and higher incomes (1), short-term (2,3) and long-term economic growth (3).

(1) Valencia Caicedo, F. (2019). The mission: Human capital transmission, economic persistence, and culture in South America. *The Quarterly Journal of Economics*, *134*(1), 507-556.

(3) Becker, S. O., & Woessmann, L. (2009). Was Weber wrong? A human capital theory of Protestant economic history. *The quarterly journal of economics*, *124*(2), 531-596.

(3) Cantoni, D., & Yuchtman, N. (2014). Medieval universities, legal institutions, and the commercial revolution. *The Quarterly Journal of Economics*, *129*(2), 823-887.

Why do governments want to intervene in education?

Financial benefits

- Study in Germany in the 1500's
- Mostly little public good provision
- Plague+ church scepticism
- Cities were more likely to adopt legislation that expanded public good provision, gave more control to secular authorities and implemented anti-corruption legislation

Source: Dittmar, J. E., & Meisenzahl, R. R. (2020). Public goods institutions, human capital, and growth: Evidence from German history. *The Review of Economic Studies*, *87*(2), 959-996.

Why do governments want to intervene in education?

Financial benefits

- Cities with public provision laws had more books, schools, and social and admin construction

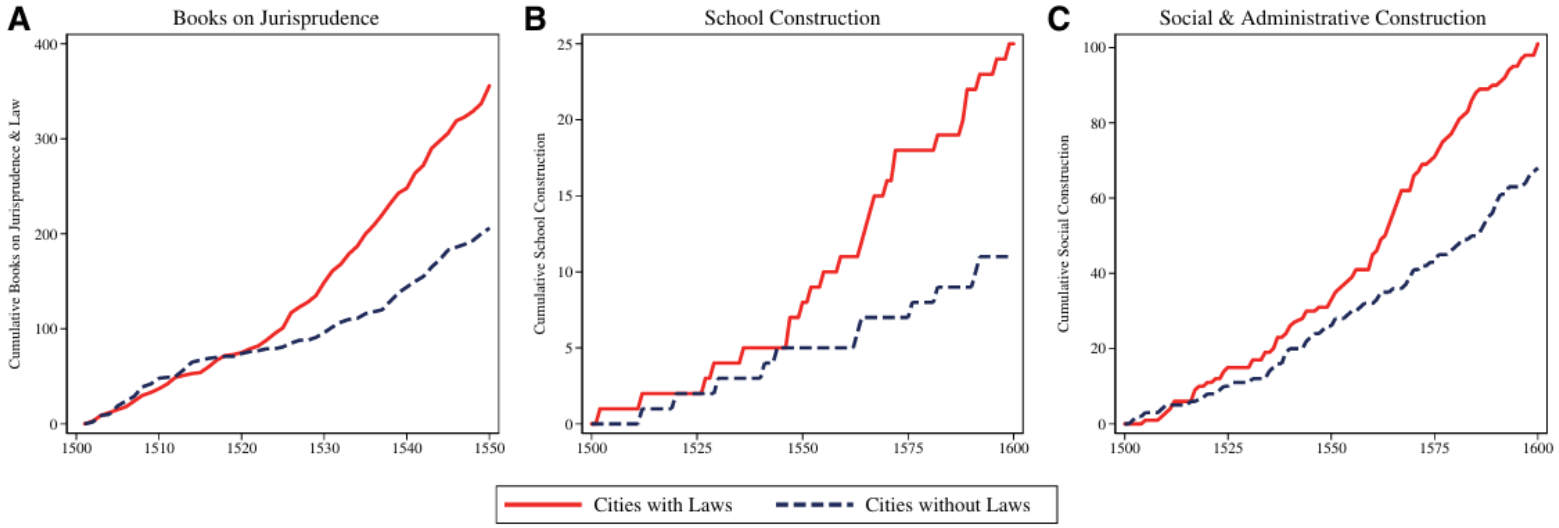

FIGURE 1

Jurisprudence and construction

Notes: This graph shows the cumulative number of (A) jurisprudence publications, (B) new school construction, and (C) administrative and social welfare construction in cities that did and did not adopt public goods institutions, measured by the passage of a church ordinance (cities with and without "Laws"). Publication and construction data are from the *Universal Short Title Catalogue* and the *Deutsches Städtebuch*, respectively, for the 239 German cities we study in this paper. We classify all new school construction as "School Construction" and all construction of buildings used by city governments and in non-education social welfare and service provision as "Social & Administrative Construction".

Why do governments want to intervene in education?

Financial benefits

- Instrument was plague which disrupted public order and discredited elites, fuelling public good law adoption (25% population dies)
- This led to public good provisions law adoption, increase in human capital and city growth

TABLE 6

IV analysis of long-run outcomes

	[1]	[2]	[3]	[4]	[5]	[6]
Panel A: first stage—public goods institutions						
	Outcome: adoption of law					
Plagues 1500–1522	0.14***	0.13***	0.13***	0.12***	0.12***	0.11***
	(0.02)	(0.02)	(0.03)	(0.02)	(0.04)	(0.03)
R^2	0.29	0.30	0.51	0.52	0.62	0.63
F Statistic on IV	34.02	32.69	18.92	31.42	8.70	12.02
Panel B: IV outcomes—population and human capital						
	Outcome: Ln Population in 1800					
Law	1.61**	1.76**	1.94**	2.17**	2.13*	2.30**
	(0.82)	(0.83)	(0.98)	(0.93)	(1.12)	(1.01)
	Outcome: Ln upper tail human capital 1750–1799					
Law	2.74**	2.96**	3.10**	3.34**	4.23**	4.54***
	(1.24)	(1.29)	(1.38)	(1.40)	(1.81)	(1.28)
	Outcome: upper tail human capital per 1,000					
Law	0.56**	0.63**	0.59*	0.66**	0.80**	0.92***
	(0.27)	(0.25)	(0.31)	(0.29)	(0.40)	(0.29)
Controls						
Baseline controls	Yes	Yes	Yes	Yes	Yes	Yes
Plagues pre-1500: level	Yes	Yes	Yes	Yes	Yes	Yes
Plagues pre-1500: non-linear	No	Yes	No	Yes	No	Yes
Region fixed effect (n=29)	No	No	Yes	Yes	No	No
Principality fixed effect (n=75)	No	No	No	No	Yes	Yes
Observations	239	239	239	239	234	234

Notes: The first-stage outcome variable in Panel A is an indicator for the adoption of a church ordinance (Law). "Plagues 1500-1522" is the number of plagues 1500–1522. The outcome variables in Panel B are: log population in 1800; log of the number of upper tail human capital individuals observed between 1750 to 1799 plus one; and the number of upper tail human capital individuals per thousand population. In first-stage regressions, the dependent variable is an indicator for the passage of a Reformation ordinance by 1600. All regressions control separately for the log of upper tail human capital observed 1420–1469 and in 1470–1519, and include the complete set of controls from Table 5. Upper tail human capital is measured by the sum of the number of migrants dying in a city-period and the number of people locally born people reaching age forty in a city-period. Region fixed effects control use territories recorded by Nüssli (2008). Principality fixed effects use principalities recorded in the *Deutsches Städtebuch*. "Plagues Pre-1500: Level" is the average number of plagues from 1350 to 1499. "Plagues Pre-1500: Non-Linear" indicates independent controls for the number of years with plague outbreaks in each of the twenty-five year periods: 1350–74 through 1475–99. Standard errors are clustered at the 1500 Region level. Statistical significance at the 90%, 95%, and 99% confidence level denoted "*", "**", and "***".

Why do governments want to intervene in education?

Financial benefits

- The adoption of public good provision adoption was related with the plague spread
- The cities that adopted those laws first experienced human capital accumulation and city growth
- Suggests that education was an important channel for development

Why do governments want to intervene in education?

Financial benefits

- Evidence that human capital (i.e., skills) generates economic growth
- Evidence that human capital has positive externalities
- There seems to be compelling evidence that it makes sense for the government to intervene in education market to assure an optimal provision level

Why do governments want to intervene in education?

Financial benefits

- It is generally accepted in economics that human capital (i.e., skills) is an input of the production function
- Growth in human capital is accepted to fuel economic growth

Why do governments want to intervene in education?

Other benefits: crime

Potential mechanisms through which education reduces crime:

- Higher human capital (education) might raise marginal returns from work as compared to common crime, although it might also raise returns from white collar crime
- Education increases patience/forward-looking
- Change social networks or peers
- More time in school reduces the time available to commit crimes (immediate effect)

Source: Hanushek, E. A., Finis Welch, Machin, S., & Ludger Woessmann. (2011). *Handbook of the economics of education*. North-Holland. Chapter 2

Why do governments want to intervene in education?

Other benefits: crime

US Sample

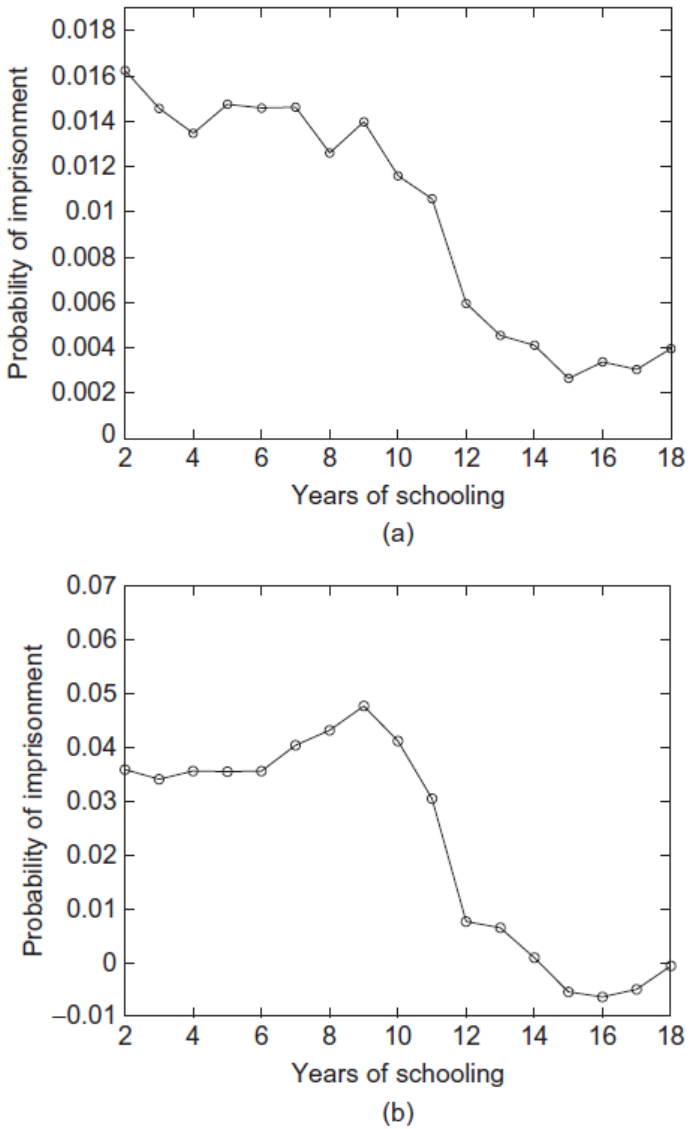

Figure 2.1 Regression-Adjusted Probability of Incarceration by Education (Men Ages 20–60). (a) Whites. (b) Blacks.

Notes: From 1960, 1970, and 1980 US censuses. Regressions control for age, state of birth, state of residence, cohort of birth, state, and year effects. Source: Lochner and Moretti (2004).

Source: Hanushek, E. A., Finis Welch, Machin, S., & Ludger Woessmann. (2011). *Handbook of the economics of education*. North-Holland. Chapter 2

Why do governments want to intervene in education?

Other benefits: crime

- US sample
- Instrument is mandatory schooling
- Different data sets (admin + self-reports)
- Negative effect on incarceration

Source: Lochner, L., & Moretti, E. (2004). The effect of education on crime: Evidence from prison inmates, arrests, and self-reports. *American economic review*, *94*(1), 155-189.

TABLE 7—IV AND CONTROL FUNCTION ESTIMATES OF THE EFFECT OF YEARS OF SCHOOLING ON IMPRISONMENT (IN PERCENTAGE TERMS)

	IV estimates			Control function
	(1)	(2)	(3)	(4)
WHITES				
Second stage				
Years of schooling	−0.11	−0.09	−0.14	−0.09
	(0.02)	(0.05)	(0.06)	(0.05)
First stage				
Compulsory attendance = 9	0.278	0.222	0.202	
	(0.026)	(0.024)	(0.024)	
Compulsory attendance = 10	0.213	0.199	0.176	
	(0.035)	(0.034)	(0.033)	
Compulsory attendance ≥ 11	0.422	0.340	0.329	
	(0.037)	(0.033)	(0.033)	
First stage F-test (d.o.f. = 3)	52.5	38.6	36.2	
Hausman test (p-value)	0.35	0.90	0.73	
Control function				
$\hat{v}$				−0.04
				(0.05)
$\hat{v}$ × years of schooling				0.00
				(0.00)
BLACKS				
Second stage				
Years of schooling	−0.47	−0.33	−0.41	−0.35
	(0.12)	(0.18)	(0.19)	(0.18)
First stage				
Compulsory attendance = 9	0.672	0.454	0.421	
	(0.043)	(0.040)	(0.039)	
Compulsory attendance = 10	0.664	0.476	0.434	
	(0.079)	(0.071)	(0.070)	
Compulsory attendance ≥ 11	0.794	0.528	0.509	
	(0.068)	(0.063)	(0.062)	
First stage F-test (d.o.f. = 3)	88.1	45.9	41.5	
Hausman test (p-value)	0.87	0.85	0.83	
Control function				
$\hat{v}$				0.20
				(0.18)
$\hat{v}$ × years of schooling				−0.02
				(0.00)
Additional controls:				
Cohort of birth effects		y	y	y
State of residence × year effects			y	

Notes: Standard errors corrected for state of birth–year of birth clustering are in parentheses. The dependent variable is a dummy equal to 1 if the respondent is in prison. Second stage and control function estimates are multiplied by 100. All specifications control for age, year, state of birth, and state of residence. See Table 4 for a description of the sample and regressors. The F-test tests whether the coefficients on the excluded instruments are jointly equal to zero. Degrees of freedom for the Hausman tests is 1.

Why do governments want to intervene in education?

Other benefits: crime

- Negative effect on arrests
- Positive externalities in terms of crime reduction

Source: Lochner, L., & Moretti, E. (2004). The effect of education on crime: Evidence from prison inmates, arrests, and self-reports. *American economic review*, *94*(1), 155-189.

TABLE 10—OLS AND IV ESTIMATES OF THE EFFECT OF SCHOOLING ON ARREST RATES

	OLS			2SLS		
	(1)	(2)	(3)	(4)	(5)	(6)
(A) AVERAGE EDUCATION						
Average years of education	−0.114	−0.116	−0.111	−0.176	−0.182	−0.162
	(0.024)	(0.023)	(0.042)	(0.080)	(0.080)	(0.105)
R^2	0.89	0.93	0.95			
(B) HIGH SCHOOL GRADUATION						
High school graduation rate	−0.618	−0.674	−0.710	−0.946	−0.941	−0.873
	(0.183)	(0.181)	(0.283)	(0.491)	(0.522)	(0.669)
R^2	0.93	0.95	0.96			
Controls:						
age × offense effects	y	y	y	y	y	y
offense × year effects	y	y	y	y	y	y
age × year effects	y	y	y	y	y	y
state × age effects	y	y	y	y	y	y
state × offense effects		y	y		y	y
state × year			y			y

Notes: Standard errors corrected for state-year-age clustering are in parentheses. The dependent variable is the logarithm of the arrest rate by age, type of offense, state, and year. Average schooling and high school graduation rate is by age group, state, and year (see text). All models control for percentage black. There are eight age groups, eight offenses, 50 states plus the District of Columbia, and four years. All models are weighted by cell size.

Why do governments want to intervene in education?

Other benefits: crime

- Instrument is increase in mandatory schooling in US
- Negative effect on jail

Table 2

Estimated Effects from an Extra Year of Compulsory Schooling

	[1]	[2]	[3]	[4]
Outcome	*Mean*	*Change in mean from 1 year of compulsory schooling before income controls*	*Change in mean from 1 year of compulsory schooling after income controls*	*Sample specification*
Log weekly income	3.06	0.131 (0.006)***	NA	Working
Log occupational prestige score	3.27	0.063 (0.003)***	0.046 (0.003)***	Working
Unemployed (*share*)	0.045	–0.005 (0.002)***	NA	Full
On welfare (*share*)	0.019	–0.015 (0.002)***	NA	Full
In jail (*share*)	0.027	–0.006 (0.003)*	NA	Black men 21–65 yrs old
In mental institute (share)	0.003	–0.001 (0.000)*	NA	Full
Probability of dying within 10 years[a]	0.11	–0.037 (0.006)***	NA	14 yrs old in 1914–1939 in 1960–1980 U.S. Censuses
First child born when teenager (*share*)	0.078	–0.008 (0.002)***	–0.008 (0.002)***	Women
Divorced (*share*)	0.137	–0.01 (0.002)***	–0.005 (0.003)***	Full
Oldest child behind grade level (*share*)	0.199	–0.032 (0.004)***	–0.026 (0.006)***	Parent matched to oldest 8–16 year old in household
Voted last national election (validated) (*share*)	0.49	0.09 (0.025)***	0.089 (0.027)***	Voting age 1978–2000 November Current Population Survey
Satisfied with life overall (*share*)	0.86	0.048 (0.010)***	0.035 (0.012)***	25–65 year olds born in UK from 1973–98 Eurobarometers

Source: Oreopoulos, P., & Salvanes, K. G. (2011). Priceless: The nonpecuniary benefits of schooling. *Journal of Economic perspectives*, *25*(1), 159-184.

Why do governments want to intervene in education?

Other benefits: crime

- England and Wales
- Instrument is increase in mandatory schooling
- Negative effect on convictions

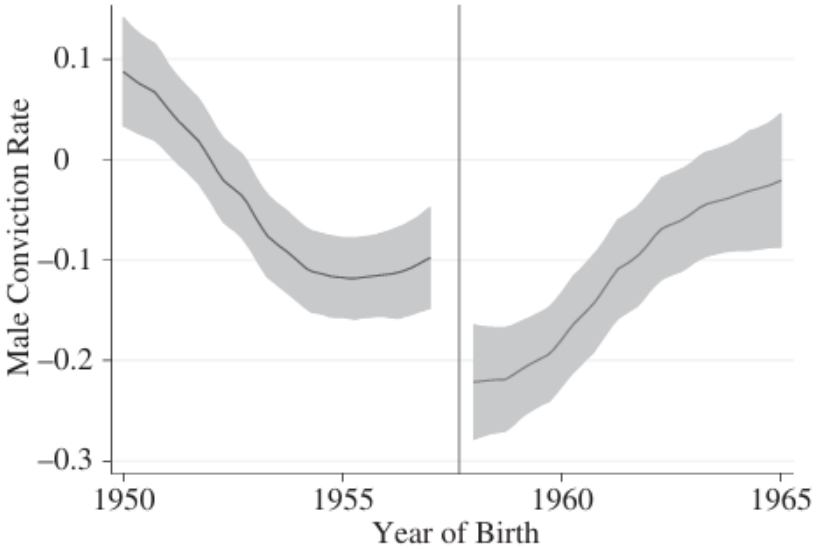

Fig. 2. *Crime Discontinuities Around the Compulsory School Leaving Age Increase*
Notes. Based on Offenders Index Data from 1972 to 1996, men aged 18–40. Graph shows residuals from regression of offence rate per 1,000 male population de-trended from a model containing GHS controls (proportion British born, proportion employed, proportion non-white and proportion living in London), year and age dummies. Lines denote kernel weighted smooth polynomial fit to data points before and after the discontinuity denoted by the vertical line. Grey shaded area is 95% confidence interval.

Source: Machin, S., Marie, O., & Vujić, S. (2011). The crime reducing effect of education. The Economic Journal, 121(552), 463-484.

Why do governments want to intervene in education?

Other benefits: crime

- England study
- Reducing Burglary Initiative (RBI): subsidies to burglary reduction schemes
- Educational Maintenance Allowances (EMA): subsidies for attendance and completion of course work
- Some areas received none, others one of the programs, some both
- Not random but authors show existence of parallel trends (Diff-in-Diff)

Source: Sabates, R., & Feinstein, L. (2008). Effects of government initiatives on youth crime. Oxford Economic Papers, 60(3), 462-483.

Why do governments want to intervene in education?

Other benefits: crime

- Reduction in burglars mainly in places that had a combination of both
- EMA and RBI seem to reduce burglary by themselves too

Source: Sabates, R., & Feinstein, L. (2008). Effects of government initiatives on youth crime. Oxford Economic Papers, 60(3), 462-483.

Table 3 Difference-in-differences in burglary for EMA, RBI, and EMA-RBI areas

	Whole sample (1)	(2)	Exclude 1998–9 (3)	Exclude 1999 (4)	Matched sample (5)
T_{EMA}*policy-on	−0.916	−0.825	−0.782	−0.746	−1.120
	(0.557)	(0.563)	(0.775)	(0.730)	(0.508)**
T_{RBI}*policy-on	−0.666	−0.531	−0.828	−0.603	—
	(0.303)**	(0.278)*	(0.426)*	(0.325)*	
$T_{EMA\text{-}RBI}$*policy-on	−1.726	−1.424	−1.590	−1.352	−1.259
	(0.615)***	(0.522)***	(0.911)*	(0.655)**	(0.521)**
Controls					
Area fixed effects	Yes	Yes	Yes	Yes	Yes
Time-varying variables	No	Yes	Yes	Yes	Yes
Yearly controls	Yes	Yes	Yes	Yes	Yes
Cohort controls	Yes	Yes	Yes	Yes	Yes
R-squared	0.51	0.53	0.55	0.54	0.65
Number of areas	132	132	132	132	18
Sample size	2,772	2,772	1,977	2,376	378

Notes: Notation follows from eq. (1), where T_{EMA} stands for LEAs where the EMA programme was piloted only, T_{RBI} for areas where the RBI programme was introduced only, and $T_{EMA\text{-}RBI}$ for areas where both initiatives were introduced. *Policy-On* indicates periods after the introduction of the initiatives.

Asterisks (*), (**), and (***) represents significant at 10, 5, and 1% levels, respectively. Estimations are weighted by population. Standard errors, in parenthesis, are clustered by LEA. Controls defined as in eq. (1).

Results hold when using the natural logarithm of convictions as outcome variable.

Why do governments want to intervene in education?

Other benefits: crime

- US sample
- Use teacher in-service days as source of variation in school attendance
- Days that teachers use for professional development, teacher conventions, planning, or parent-teacher conferences

Source: Jacob, B. A., & Lefgren, L. (2003). Are idle hands the devil's workshop? Incapacitation, concentration, and juvenile crime. *American economic review*, *93*(5), 1560-1577.

Why do governments want to intervene in education?

Other benefits: crime

- Teacher in-service days increase property crimes
- Teacher in-service days decrease violent crimes
- School increases the level of interaction among adolescents, raising the likelihood of violent conflicts

Source: Jacob, B. A., & Lefgren, L. (2003). Are idle hands the devil's workshop? Incapacitation, concentration, and juvenile crime. *American economic review*, *93*(5), 1560-1577.

TABLE 3—THE RELATIONSHIP BETWEEN REPORTED JUVENILE OFFENDERS AND THE SCHOOL TERM

	(Dependent variable = number of reported juvenile offenders for the following offenses)				
	Serious offenses				Minor offenses
Independent variables	All crimes	Property crimes	Violent crimes	Other crimes	All crimes
Teacher in-service day	−0.007	0.133	−0.323	−0.106	−0.153
	(0.16)	(2.78)	(4.40)	(1.07)	(2.57)
Idiosyncratic break	0.032	0.154	−0.204	−0.089	−0.079
	(1.02)	(4.17)	(3.94)	(1.13)	(1.77)
Common break	−0.112	0.021	−0.478	−0.353	−0.271
	(4.31)	(0.71)	(10.00)	(5.18)	(6.90)
Holiday	−0.268	−0.179	−0.418	−0.343	−0.388
	(10.63)	(6.05)	(9.85)	(5.47)	(10.30)
Summer vacation	−0.018	0.068	−0.220	−0.105	−0.128
	(0.89)	(2.90)	(6.52)	(2.60)	(4.33)
Half-day	0.172	0.156	0.101	0.159	−0.126
	(3.27)	(2.36)	(1.28)	(1.20)	(1.43)
Tuesday	−0.007	−0.039	0.046	0.078	0.073
	(0.52)	(2.29)	(2.15)	(2.29)	(3.69)
Wednesday	−0.028	−0.060	0.045	0.031	0.043
	(2.03)	(3.54)	(2.07)	(0.89)	(2.15)
Thursday	−0.040	−0.061	−0.011	0.038	0.023
	(2.84)	(3.60)	(0.52)	(1.12)	(1.14)
Friday	0.054	0.019	0.052	0.206	0.170
	(3.94)	(1.16)	(2.39)	(6.17)	(8.71)
Saturday	−0.071	0.029	−0.374	−0.108	−0.052
	(5.06)	(1.74)	(15.60)	(3.05)	(2.55)
Sunday	−0.294	−0.243	−0.416	−0.302	−0.212
	(19.69)	(13.65)	(17.15)	(8.09)	(9.97)
City * month * year and day-of-week fixed effects	Yes	Yes	Yes	Yes	Yes
Observations	38,339	38,339	38,308	37,181	37,451

Notes: The estimates come from negative binomial regressions. *Z*-statistics are included in parentheses. Serious offenses are defined as NIBRS type A offenses and include primarily property, violent, and drug crimes. Minor offenses are defined as NIBRS type B offenses and include offenses such as trespassing, loitering, disorderly conduct, and liquor law violations. National holidays include days such as Thanksgiving, Christmas, New Year's Day, Easter, President's Day, and Memorial Day. Idiosyncratic breaks are days that a school district has off and no more than 30 percent of all school districts have off. Common breaks are days that more than 30 percent of schools have off. Teacher in-service days are days during the regular school year that students do not attend school and districts reserve for teachers to attend professional development workshops or school-based planning activities.

Why do governments want to intervene in education?

Other benefits: crime

- US sample
- Increase in mandatory schooling on contemporaneous crime
- Decrease in property crime and violent crime

Source: Anderson, D. M. (2014). In school and out of trouble? The minimum dropout age and juvenile crime. *Review of Economics and Statistics*, *96*(2), 318-331.

Why do governments want to intervene in education?

Other benefits: crime

TABLE 4.—TEEN ARREST RATES AND THE MINIMUM DROPOUT AGE, 1980–2008

	All Crimes			Property Crime			Violent Crime			Drug Crime		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Males												
MDA18	−10.20*	−10.20*	−10.27*	−3.270**	−3.232**	−3.241**	−3.262**	−3.268**	−3.264**	−3.287	−3.275	−3.309
×*age16to18*	(5.331)	(5.321)	(5.323)	(1.303)	(1.296)	(1.295)	(1.468)	(1.466)	(1.468)	(3.431)	(3.426)	(3.436)
N	134,296	134,293	134,293	134,037	134,034	134,034	134,116	134,114	134,114	133,860	133,857	133,857
R^2	0.830	0.835	0.841	0.742	0.762	0.779	0.868	0.872	0.877	0.619	0.634	0.643
Females												
MDA18	−1.419	−1.448	−1.553	−0.739	−0.740	−0.749	−0.634**	−0.643**	−0.656**	0.029	0.030	0.027
×*age16to18*	(1.227)	(1.203)	(1.192)	(0.837)	(0.835)	(0.833)	(0.303)	(0.304)	(0.304)	(0.439)	(0.438)	(0.439)
N	134,307	134,304	134,304	133,727	133,725	133,725	133,860	133,857	133,857	133,352	133,349	133,349
R^2	0.747	0.756	0.773	0.709	0.718	0.733	0.717	0.727	0.746	0.586	0.599	0.611
County FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Covariates	No	Yes	Yes	No	Yes	Yes	No	Yes	Yes	No	Yes	Yes
State trends	No	No	Yes	No	No	Yes	No	No	Yes	No	No	Yes

Each cell represents a separate regression. The dependent variable is the arrest rate per 1,000 of the relevant age group population. The control group consists of individuals 13 to 15 years of age. Other covariates control for county demographic variables, population density, the minimum wage, the minimum legal drinking age, and the count of agencies reporting each year. County mean populations are used as weights. Standard errors are corrected for clustering at the state level. Significant at *10%, **5%, ***1%.

Source: Anderson, D. M. (2014). In school and out of trouble? The minimum dropout age and juvenile crime. *Review of Economics and Statistics*, *96*(2), 318-331.

Why do governments want to intervene in education?

Other benefits: crime

- US sample (Colorado)
- Implementation of a 4-day school week
- Increase in property crime, especially outside of school

Source: Fischer, Stefanie, and Daniel Argyle. "Juvenile crime and the four-day school week." *Economics of education Review* 64 (2018): 31-39.

Table 2
Effect of four-day school week on crime and location.

	Crime type				Location	
	All crime (1)	Property (2)	Drugs (3)	Violent (4)	Not-school (5)	At-school (6)
Panel A: Agency, year fixed effects						
Four-day	0.255***	0.315***	0.237	0.201	0.229***	0.274*
	(0.084)	(0.089)	(0.146)	(0.167)	(0.088)	(0.160)
Panel B: Agency, year fixed effects and time-varying controls						
Four-day	0.196**	0.266***	0.252	0.076	0.182**	0.175
	(0.086)	(0.090)	(0.167)	(0.124)	(0.092)	(0.164)
N	836	825	763	808	836	728

Notes: Each coefficient comes from a separate regression. Each regression in Panel A includes year and agency fixed effects and each regression in Panel B includes year and agency fixed effects, and agency-level time-varying controls (fraction eligible for free or reduced lunch, student-teacher ratio, unemployment rate, and fraction white). Each regression also includes total enrollment as the exposure variable with coefficient constrained to be one. Observations vary across outcomes because, in some agencies for some crime types and locations, zero crime counts are observed in all sample years. Standard errors are clustered at the agency level and are reported in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Why do governments want to intervene in education?

Other benefits: crime

- Evidence that increases in education reduce long-term crime
- Effect on contemporaneous crime is mixed, school decreases property crime, but has the potential to increase violent crime, through network and peer effects
- Research tends to focus on common crime
- Less is known about the effects of education on white-collar crime

Why do governments want to intervene in education?

Other benefits: health and welfare

- Last class we found positive effects on health
- Evidence of negative effects of education on receiving welfare as support

Source: Oreopoulos, P., & Salvanes, K. G. (2011). Priceless: The nonpecuniary benefits of schooling. *Journal of Economic perspectives*, *25*(1), 159-184.

Table 2

Estimated Effects from an Extra Year of Compulsory Schooling

	[1]	[2]	[3]	[4]
Outcome	*Mean*	*Change in mean from 1 year of compulsory schooling before income controls*	*Change in mean from 1 year of compulsory schooling after income controls*	*Sample specification*
Log weekly income	3.06	0.131 (0.006)***	NA	Working
Log occupational prestige score	3.27	0.063 (0.003)***	0.046 (0.003)***	Working
Unemployed (*share*)	0.045	–0.005 (0.002)***	NA	Full
On welfare (*share*)	0.019	–0.015 (0.002)***	NA	Full
In jail (*share*)	0.027	–0.006 (0.003)*	NA	Black men 21–65 yrs old
In mental institute (share)	0.003	–0.001 (0.000)*	NA	Full
Probability of dying within 10 years[a]	0.11	–0.037 (0.006)***	NA	14 yrs old in 1914–1939 in 1960–1980 U.S. Censuses
First child born when teenager (*share*)	0.078	–0.008 (0.002)***	–0.008 (0.002)***	Women
Divorced (*share*)	0.137	–0.01 (0.002)***	–0.005 (0.003)***	Full
Oldest child behind grade level (*share*)	0.199	–0.032 (0.004)***	–0.026 (0.006)***	Parent matched to oldest 8–16 year old in household
Voted last national election (validated) (*share*)	0.49	0.09 (0.025)***	0.089 (0.027)***	Voting age 1978–2000 November Current Population Survey
Satisfied with life overall (*share*)	0.86	0.048 (0.010)***	0.035 (0.012)***	25–65 year olds born in UK from 1973–98 Eurobarometers

Why do governments want to intervene in education?

Evidence supports the idea that education:

- Generates economic growth (above individual returns)
- Reduces crime
- Improves health
- Decreases reliance on welfare
- ….
- Compelling evidence that it improves well-being of population
- Market failure that government should address

Why do governments want to intervene in education?

Next class:

The effects of education on civic engagement

What happens when we are in the presence of a non-benevolent/autocratic government?